Question 1

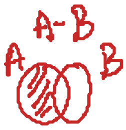
a) Let the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Sets A, B and C are all subsets of the universal set U, defined as follows:

$$A = \{x \mid x \text{ is an odd number}\};$$
$$B = \{x \mid x \text{ is multiple of } 3\};$$
$$C = \{x \mid x \text{ is a factor of } 18\}.$$

(i) List all elements of sets A, B and C. (3 marks)

(ii) Find $A \cap B$. (1 mark)

(iii) Find $A - B, B - A$, and $A \oplus B$. (3 marks)



(iv) Find $\overline{A \cup C}$. $= \bar{A} \cap \bar{C}$ (2 marks)

(i) $A = \{1, 3, 5, 7, 9\}$

$B = \{3, 6, 9\}$

$C = \{1, 2, 3, 6, 9\}$

(ii) $A \cap B = \{3, 9\}$

(iii) $A - B = \{1, 5, 7\}$

$B - A = \{6\}$

$A \oplus B = \{1, 5, 6, 7\}$

(iv) $A \cup C = \{1, 2, 3, 5, 6, 7, 9\}$

$\overline{A \cup C} = \{4, 8, 10\}$

b) Given $a = 2020$ and $b = 92$.

(i) Use the Euclidean algorithm to find the Greatest Common Divisor (GCD) of a and b. (3 marks)

(ii) Present the Greatest Common Divisor (GCD) a and b in the form of $sa + tb$ where $s, t \in \mathbb{Z}$. (3 marks)

(iii) Hence, compute the Least Common Multiple (LCM) of a and b. (2 marks)

(i) $2020 = 92(21) + 88$

$92 = 88(1) + 4$

$88 = 4(22) + 0$

$\gcd(2020, 92) = 4$

$88 = 2020 - 92(21)$

$4 = 92 - 88$

(ii) $4 = 92 - 88$

$= 92 - [2020 - 92(21)]$

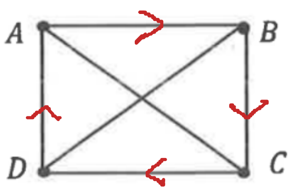
$= 92 - 2020 + 92(21)$

$= (22)92 + (-1)2020$

(iii) $\text{lcm}(2020, 92) = \dfrac{2020 \times 92}{4}$

$= 46460$

c) Consider the graph below:



(i) Write the degree of each vertex of the graph. (2 marks)

(ii) Identify whether the graph contains a Euler circuit. If it does, write down a Euler circuit by listing the vertices in the order they are visited, starting and ending at vertex A. If it does not, explain why. (2 marks)

(iii) Identify whether the graph contains a Euler path. If it does, write down a Euler path by listing the vertices in the order they are visited, starting at vertex A. If it does not, explain why. (2 marks)

(iv) Identify whether the graph has a Hamiltonian circuit? If it does, write down a Hamiltonian circuit by listing the vertices in the order they are visited, starting and ending at vertex A. If it does not, explain why. (2 marks)

(i)

vertex	A	B	C	D
degree	3	3	3	3

(ii) No Euler circuit because not all vertices have even degrees

(iii) No Euler path because it does not have exactly two vertices with odd degrees.

(iv) Yes, Hamiltonian circuit exist.

Hamiltonian circuit: A, B, C, D, A

Question 2

a) Let $A = \{1, 2, 3, 4\}$ and let the relation R on the set A be defined by

$$R = \{(1,1), (1,2), (1,4), (2,1), (2,2), (2,4), (3,4), (4,1), (4,2), (4,4)\}.$$

(i) Construct the matrix representation of R. (2 marks)

(ii) Compute the in-degree and out-degree of each vertex. (4 marks)

(iii) Draw the digraph of the relation R. (2 marks)

(iv) Examine whether the relation R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric and transitive. Give a counterexample if the answer is 'No'. (7 marks)

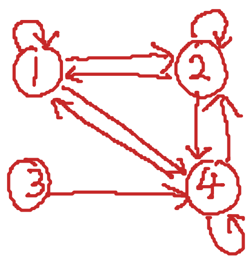
(v) Identify whether R is an equivalence relation on A? Justify your answer. (2 marks)

(i) $M_R = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

(ii)

	vertex	1	2	3	4
(col)	in-degree	3	3	0	4
(row)	out-degree	3	3	1	3

(iii)



(iv) R is not reflexive since $(3,3) \notin R$

R is not irreflexive since $(1,1) \in R$

R is not symmetric since $(3,4) \in R$ but $(4,3) \notin R$

R is not asymmetric since $(1,1) \in R$

R is not antisymmetric since $(1,2) \in R$ and $(2,1) \in R$ but $1 \neq 2$

R is not transitive since $(3,4) \in R$ and $(4,1) \in R$ but $(3,1) \notin R$

(v) R is not an equivalence relation since R is not reflexive, not symmetric and not transitive.

b) Let $A = \{1, 2, 3, 4\}$ and the relation R on the set A be represented by the matrix

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

$$\Delta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(i) Find the matrix of the reflexive closure of R. (1 mark)

(ii) Find the matrix of the symmetric closure of R. (2 marks)

(iii) Apply Warshall's algorithm to obtain the matrix of the transitive closure of R. M_{R^∞} (5 marks)

(i) $M_{R\cup\Delta} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

$1 \vee 1 = 1$
$1 \vee 0 = 1$
$0 \vee 1 = 1$
$0 \vee 0 = 0$

(ii) $M_{R\cup R^{-1}} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

(iii) Let $W_0 = M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

C_1: 1, 2
R_1: 1, 3
Add: (1,1), (1,3), (2,1), (2,3)

$W_1 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

C_2: 4
R_2: 1, 3, 4
Add: (4,1), (4,3), (4,4)

$W_2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

C_3: 1, 2, 3, 4
R_3: 3
Add: (1,3), (2,3), (3,3), (4,3)

$$W_3 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

C_4: 2, 4

R_4: 1, 2, 3, 4

Add: (2,1), (2,2), (2,3), (2,4),
(4,1), (4,2), (4,3), (4,4)

$$W_4 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} = M_{R^\infty}$$

Question 3

a) Let $A = \{1, 2, 3, 4\}$ and let the relations R and S on A be given in matrix form below

$$M_R = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}; \; M_S = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

(i) Compute $M_{\overline{R \cap S}}$, $\quad 1 \cap 1 = 1 \quad 0 \cap 1 = 0$ $\quad 1 \cap 0 = 0 \quad 0 \cap 0 = 0$ (2 marks)

(ii) Compute $M_{(R \cup S)^{-1}}$, $\quad 1 \cup 1 = 1 \quad 0 \cup 1 = 1$ $\quad 1 \cup 0 = 1 \quad 0 \cup 0 = 0$ (2 marks)

(iii) Compute $M_{\overline{S \circ R}}$. (3 marks)

(i) $$M_{\overline{R \cap S}} = \overline{\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

(ii) $$M_{(R \cup S)^{-1}} = \left(\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(iii) $$M_{S \circ R} = M_R \odot M_S$$

$$= \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$M_{\overline{S \circ R}} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

b) Given $A = \{1, 2, 3, 4, 5\}$. Let $\rho_1 = (1, 3, 5)$ and $\rho_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 5 & 2 \end{pmatrix}$ be permutations on the set A.

(i) Compute ρ_1^{-1} and $(\rho_1^{-1} \circ \rho_2)$. (4 marks)

(ii) Write $(\rho_1^{-1} \circ \rho_2)$ as a product of disjoint cycles, and also as a product of transpositions. (2 marks)

(iii) Determine whether permutation $(\rho_1^{-1} \circ \rho_2)$ is even or odd. Justify your answer. (2 marks)

(i) $P_1^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 4 & 3 \end{pmatrix}$

$$P_1^{-1} \circ P_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 4 & 3 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 5 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 5 & 3 & 2 \end{pmatrix}$$

(ii) $P_1^{-1} \circ P_2 = (2, 4, 3, 5)$

$= (2, 5) \circ (2, 3) \circ (2, 4)$

(iii) Odd permutation since number of transpositions is 3

c) Consider the (2, 5) encoding function $e: B^2 \to B^5$ defined by

$0 \oplus 0 = 0$
$1 \oplus 0 = 1$
$0 \oplus 1 = 1$
$1 \oplus 1 = 0$

$e(00) = 00000, = C_0$ $\quad$ $e(10) = 10101, = C_2$
$e(01) = 01111, = C_1$ $\quad$ $e(11) = 11010. = C_3$

$C_1 \oplus C_2$: 01111 $\oplus$ 10101 = 11010

$C_1 \oplus C_3$: 01111 $\oplus$ 11010 = 10101

$C_2 \oplus C_3$: 10101 $\oplus$ 11010 = 01111

(i) Let $G = \{C_o, C_1, C_2, C_3\}$ where $C_o = 00000$, $C_1 = 01111$, $C_2 = 10101$, and $C_3 = 11010$. Complete the table below. (4 marks)

$\oplus$	$C_o = 00000$	$C_1 = 01111$	$C_2 = 10101$	$C_3 = 11010$
$C_o = 00000$	$C_0 = 00000$	$C_1 = 01111$	$C_2 = 10101$	$C_3 = 11010$
$C_1 = 01111$	$C_1 = 01111$	$C_0 = 00000$	$C_3 = 11010$	$C_2 = 10101$
$C_2 = 10101$	$C_2 = 10101$	$C_3 = 11010$	$C_0 = 00000$	$C_1 = 01111$
$C_3 = 11010$	$C_3 = 11010$	$C_2 = 10101$	$C_1 = 01111$	$C_0 = 00000$

(ii) Verify that the (2,5) encoding function $e: B^2 \to B^5$ is a group code. (4 marks)

(iii) Compute the minimum distance of the group code, and determine how many errors the encoding function e can detect. (2 marks)

(ii) $C_i \oplus C_j \in G$ for $i = 0, 1, 2, 3$ and $j = 0, 1, 2, 3$

The identity is $C_0 = 00000 \in G$

$C_0^{-1} = C_0 \in G$

$C_1^{-1} = C_1 \in G$

$C_2^{-1} = C_2 \in G$

$C_3^{-1} = C_3 \in G$

$\therefore$ It is a group code.

(iii)

codeword	$C_1 = 01111$	$C_2 = 10101$	$C_3 = 11010$
weight	4	3	3

minimum distance = 3

Can detect 2 or fewer error

Question 4

a) Let $A \equiv \sim(p \longrightarrow q) \leftrightarrow (q \wedge r)$.

(i) Construct a truth table for the expression A. Your truth table must be in the following form.

p	q	r	$(p \longrightarrow q)$	$\sim(p \longrightarrow q)$	$(q \wedge r)$	$\sim(p \longrightarrow q) \leftrightarrow (q \wedge r)$
T	T	T	T	F	T	F
T	T	F	T	F	F	T
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	F	T	F
F	T	F	T	F	F	T
F	F	T	T	F	F	T
F	F	F	T	F	F	T

(4 marks)

(ii) Determine whether the expression $A \equiv \sim(p \longrightarrow q) \leftrightarrow (q \wedge r)$ is a tautology (All T), contradiction (All F) or contingency (T/F). (1 mark)

(iii) Obtain the Principal Disjunctive Normal Form (PDNF) and the Principal Conjunctive Normal Form (PCNF) of A. Hence, deduce the PDNF and PCNF of $\sim A$. (7 marks)

(ii) contingency

(iii) PDNF of $A = pq\bar{r} + \bar{p}q\bar{r} + \bar{p}\bar{q}r + \bar{p}\bar{q}\bar{r}$

PDNF of $\sim A = pqr + p\bar{q}r + p\bar{q}\bar{r} + \bar{p}qr$

PCNF of $A = \sim$(PDNF of $\sim A$)

$= (\bar{p}+\bar{q}+\bar{r})(\bar{p}+q+\bar{r})(\bar{p}+q+r)(p+\bar{q}+\bar{r})$

PCNF of $\sim A = \sim$(PDNF of A)

$= (\bar{p}+\bar{q}+r)(p+\bar{q}+r)(p+q+\bar{r})(p+q+r)$

b) Simplify and show that $[(x \vee y')' \vee (x' \wedge y')] \vee (x' \wedge z) = x'$ using the laws of Boolean algebra.

(6 marks)

$[(x \vee y')' \vee (x' \wedge y')] \vee (x' \wedge z)$

$= [(x' \wedge y) \vee (x' \wedge y')] \vee (x' \wedge z)$

$= [x' \wedge (y \vee y')] \vee (x' \wedge z)$

$= (x' \wedge 1) \vee (x' \wedge z)$

$= x' \wedge (1 \vee z)$

$= x' \wedge 1$

$= x'$

c) Let $f(x, y, z, w) = (x' \wedge y' \wedge z \wedge w') \vee (x \wedge y' \wedge z' \wedge w) \vee (x' \wedge y \wedge z \wedge w') \vee (x' \wedge y' \wedge z \wedge w) \vee (x' \wedge y \wedge z \wedge w) \vee (x \wedge y \wedge z' \wedge w)$.

(i) Construct a Karnaugh map using the form given below. (3 marks)

	z'	z'	z	z	
x'			1	1	y'
x'			1	1	y
x		1			y
x		1			y'
	w'	w	w	w'	

(ii) Then, simplify $f(x, y, z, w)$ to its simplest form using the Karnaugh map. (4 marks)

$f(x,y,z,w) = x'z + xwz'$